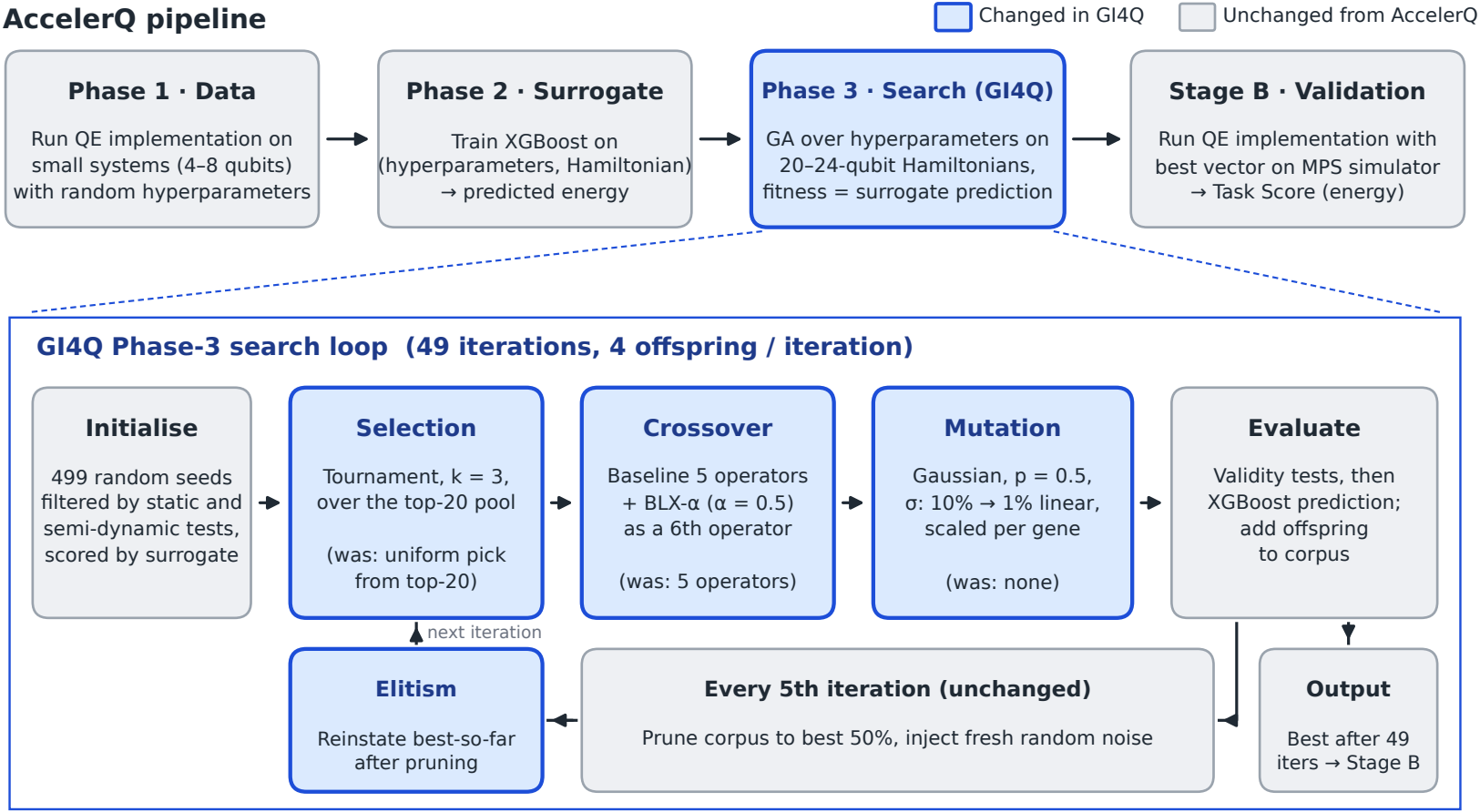
AccelerQ pipeline
Changed in GI4Q
Unchanged from AccelerQ
Phase 1 · Data
Run QE implementation on small systems (4–8 qubits) with random hyperparameters
Phase 2 · Surrogate
Train XGBoost on (hyperparameters, Hamiltonian) → predicted energy
Phase 3 · Search (GI4Q)
GA over hyperparameters on 20–24-qubit Hamiltonians, fitness = surrogate prediction
Stage B · Validation
Run QE implementation with best vector on MPS simulator → Task Score (energy)
GI4Q Phase-3 search loop (49 iterations, 4 offspring / iteration)
Initialise
499 random seeds filtered by static and semi-dynamic tests, scored by surrogate
Selection
Tournament, k = 3, over the top-20 pool
(was: uniform pick from top-20)
Crossover
Baseline 5 operators + BLX-α (α = 0.5) as a 6th operator
(was: 5 operators)
Mutation
Gaussian, p = 0.5, σ: 10% → 1% linear, scaled per gene
(was: none)
Evaluate
Validity tests, then XGBoost prediction; add offspring to corpus
next iteration
Elitism
Reinstate best-so-far after pruning
Every 5th iteration (unchanged)
Prune corpus to best 50%, inject fresh random noise
Output
Best after 49 iters → Stage B